Distilling Task Knowledge or Model-Specific Patches? A Cross-Model Diagnostic for Failure-based Textual Distillation

Anonymous Author(s)
Affiliation
Address
email

Abstract

CS-ICL turns task demonstrations into a reusable cheatsheet with a single LLM pass. We ask whether more task-related knowledge can be added to such a cheatsheet under iterative refinement using the mistakes of one source model. It turns out that this approach is not always successful and the main reason is simple: the source and target models may have different failure pattern. Rules distilled from source model failures can repair the source model while confusing other models, which can be examined by cheatsheet transferability. We show that failure-driven refinement can encode model-specific reasoning patches rather than transferable task knowledge. To identify and diagnose this phenomenon, we utilize Failure-set overlap as a pre-refinement risk screen, while identifying oracle contamination and partition overfitting for explaining how source-specificity enters the artifact. Finally we develop an oracle-free holistic rewrite pipeline that reduces these pressures and improves cross-family transfer on shared-failure tasks. Evaluated across five tasks (three BBH, two AGIEval) on a single seed, the pipeline achieves consistent positive transfer cross model families. We argue that failure-driven iterative refinement helps knowledge distillation only when source and target failures substantially overlap and oracle-free holistic rewrites are used in place of partition-specific memory atoms.

1 Introduction

Large language models can distill task demonstrations into compact natural-language artifacts that are prepended to every input and reused across models. A *cheatsheet* [Honda et al., 2025] is one such artifact: a single-pass ruleset that provides a competitive baseline at a fraction of many-shot ICL's per-input token cost. These artifacts are useful only if they contain knowledge that survives beyond the source model that generated them.

Failure-driven augmentation sharpens a fundamental trade-off: optimizing against source-model failures can improve the source while encoding either transferable task structure or model-specific patches. We test which side dominates by evaluating refined cheatsheets on non-source model families. To understand how vulnerable are cheatsheet contents to encode model-specific reasoning pattern structurally, We use ICREFINE, a three-phase iterative cheatsheet refinement system, as a controlled testbed for when failure-driven distillation encodes transferable knowledge or source-specific patches. We also use BBH as the primary diagnostic setting where bifurcations and mechanism ablations are clearest with the three AIGEval tasks to make sure our result generalize beyond just BBH tasks.

Our empirical finding is not simply that refinement fails to transfer. The clearest symptom is a source-gain/target-loss bifurcation: on `geometric_shapes`, ICREFINE Phase 2 case studies improve the source model by $+5.0$ pp while regressing Claude by -3.3 pp and Llama by -6.7 pp. We further

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

show that this behavior depends on source–target failure overlap, is amplified by oracle contamination and partition overfitting, and can be mitigated by oracle-free holistic rewrites. The central question is therefore not whether refinement adds information, but whether it selects task knowledge or model-specific patches.

Contributions. (1) **Diagnostic framework.** We operationalize task-related content vs. source-specific patches using shared and source-private failure regions and a transfer threshold. (2) **Cross-model transfer evidence.** We show that failure-based distillation can improve the source model while regressing targets, with BBH providing the clearest source-gain/target-loss bifurcation and AGIEval stress-testing the framework across different accuracy baseline. (3) **Failure-mode identification.** We identify oracle contamination as the primary and most counterintuitive mechanism: gold CoT can encode source-family reasoning traces rather than model-independent task knowledge when fed into cheatsheet refinement process. We further show that partition overfitting and bootstrap convergence amplify this specificity pressure within ICREFINE. (4) **Holistic v7 pipeline.** Applying the failure-mode analysis directly, we develop holistic v7: a failured based cheatsheet refinement pipeline based on CS-ICL initiation with no oracle injection, no partition-based case studies, and early stopping. We show that it achieves consistent positive transfer on tasks with structurally shared failure modes ($+8$–$+23$ pp on `geometric_shapes`, $+1$–$+4$ pp on `causal_judgement`) while avoiding the regressions induced by ICREFINE's oracle-injected partition-based design.

2 Preliminaries: Textual Distillation and Transfer Diagnostics

2.1 Distillation Setting, Representativeness, and Transfer Threshold

We consider multi-class classification tasks where a *cheatsheet* $\mathcal{C} = (\mathrm{PK}, \mathcal{S})$ consists of a prior-knowledge block PK (free-form ruleset, $\leq$12 K characters) and optional distilled knowledge atoms $\mathcal{S}$ (case studies in an ACTIVATE IF / WHY format). A static cheatsheet with $|\mathcal{S}| = 0$ is the CS-ICL baseline [Honda et al., 2025]. The source model M_{src} performs all distillation roles; target models $\{M_{\mathrm{tgt}}\}$ only use the cheatsheet to aid their prediction at inference time. A cheatsheet *transfers* if it improves M_{tgt} above CS-ICL when $M_{\mathrm{tgt}} \neq M_{\mathrm{src}}$.

Let $F_s = F_{M_s}(\mathcal{C}_0)$ and $F_t = F_{M_t}(\mathcal{C}_0)$ denote the CS-ICL baseline failure sets. We call $F_s \cap F_t$ the *shared-failure region* and $F_s \setminus F_t$ the *source-private region*. The key governing quantity is *source-failure representativeness*:

$$\rho_{s\to t} = \frac{|F_s \cap F_t|}{|F_s|},$$

the fraction of source failures that are also target failures. We use $\rho_{s\to t}$ as an operational measure of whether source failures are representative of target failures: high values make transfer structurally possible, while low values indicate source-specificity risk.

A minimal formal lens separates three quantities that otherwise conflated with one another in transfer analysis. Let $a_c(x) \in \{0, 1\}$ be the activation predicate of knowledge atom c, with activation precision $\pi_M(c) = \Pr[x \in F_M \mid a_c(x) = 1]$, fix rate $f_M(c)$, and regression rate $r_M(c)$. A correction c has positive marginal effect on model M if and only if

$$\pi_M(c) > \frac{r_M(c)}{f_M(c) + r_M(c)}. \tag{1}$$

The same correction may satisfy Eq. (1) for the source model ($\pi_{M_{\mathrm{src}}}$ high) while violating it for the target ($\pi_{M_{\mathrm{tgt}}}$ low), if the activation predicate fires on inputs the target already answers correctly. Full derivation and gate thresholds are in Appendix A.

2.2 Failure-Set Jaccard as a Specificity-Risk Screen

We define pairwise failure-set Jaccard on the CS-ICL baseline:

$$J(M_1, M_2; T) = \frac{|F_{M_1}(\mathcal{C}_0) \cap F_{M_2}(\mathcal{C}_0)|}{|F_{M_1}(\mathcal{C}_0) \cup F_{M_2}(\mathcal{C}_0)|}.$$

For a correction generated from source model failures with $A_c \subseteq F_s$, target activation precision satisfies $\pi_{M_{\mathrm{tgt}}}(c) \leq |F_s \cap F_t|/|A_c|$. In the idealized case where the learned predicate broadly covers

Component	Distillation role	Specificity pressure
Phase 0 bootstrap	Generate initial prior knowledge from 75 training examples in a single pass	Low—holistic task-level compression
Phase 1 PK patching	Iteratively revise the prior-knowledge block from source-model failures using fix/regression gates	Moderate—source-failure patches to PK
Phase 2 case studies	Partition remaining failures and generate ACTIVATE IF / WHY knowledge atoms	High—localized source-failure memory atoms
Holistic v7	CS-ICL warmstart with iterative whole-cheatsheet rewrites; no oracle injection, no case studies, early stopping	Low—global rewrite avoids partition-specific memory atoms

Table 1: ICREFINE components and specificity pressure each induces. Higher pressure means distilled content is more likely to encode source-specific patches; lower pressure favors task-related knowledge. Full algorithmic details in Appendix A.

Table 2: Diagnostic interventions and their identification logic. Each intervention is paired with a prediction: the stated outcome supports the mechanism if observed.

Intervention	Mechanism tested	Identification logic
Case-study contrib. Δ_{cs}	Partition overfitting	If Phase 2 helps source but hurts targets, local memory atoms encode source-private failures
No-oracle (E3)	Oracle contamination	If removing gold CoT improves transfer, oracle traces were over-specific rather than task-level rules
EA Phase 1	Bootstrap convergence	If better holistic PK removes need for case studies, premature PK convergence caused patching
Holistic v7	Conservative control	If oracle-free holistic rewrites improve transfer, specificity pressure, not task difficulty, is the bottleneck
ProTeGi	Optimizer-side check	If the same transfer asymmetry appears under a different optimizer, it is not ICREFINE-specific

the source-failure region ($A_c \approx F_s$), this approaches $\rho_{s\to t}$. Because real predicates may cover subsets or supersets of Fs, Jaccard is only a one-sided risk screen: low J warns against refinement, while high J does not guarantee gains.

3 Methods: Distillation Pipelines and Diagnostic Interventions

3.1 ICREFINE

We established ICREFINE as our primary diagnostic pipeline. Phase 0 produces a holistic prior-knowledge block from 75 training examples in a single pass. Phase 1 iteratively rewrites that block using source-model failures, subject to a fix-rate and a regression gate. Phase 2 converts remaining failure partitions into localized ACTIVATE IF / WHY case studies through mechanical failed case partitioning that partition wrongly-predicted cases into different bins based on their structural similarities. Each stage risks encoding more source-model-specific content than the last: Phase 0 compresses task-level signal, Phase 1 introduces source-failure-conditioned edits, and Phase 2 stores local source-failure memories targeted at certain type of questions. Table 1 maps each component to this risk; the experiments in Sections 5–8 use these components as probes.

3.2 Diagnostic Interventions

We identify knowledge distillation failure mechanisms through targeted interventions, each paired with a falsifiable prediction. Table 2 lists each intervention, the mechanism it probes, and the identification logic.

3.3 Holistic v7 and ProTeGi

Holistic v7. Holistic v7 is the conservative design alternative we evaluate against ICREFINE. It warm-starts from a CS-ICL cheatsheet generated by gpt-4.1-mini (1000-token cap), then iterates whole-cheatsheet rewrites without oracle CoT injection and without partition-based case studies. Candidate revisions are accepted via a fix-rate gate and regression gate; the best-so-far cheatsheet is tracked with rollback, and early stopping triggers after three consecutive iterations with no improvement. The design deliberately avoids oracle contamination and partition overfitting, at the cost of requiring more iterations to converge.

ProTeGi. To test whether the trade-off is specific to ICREFINE, we run ProTeGi [Pryzant et al., 2023] under the same cross-model transfer criterion. ProTeGi is initialized from CS-ICL, optimized with gpt-4.1-mini as the training model, and evaluated on non-source models without further adaptation. Full optimizer settings are in Appendix G.

4 Experimental Setup

Datasets and models. We evaluate on 11 BBH tasks [Suzgun et al., 2022]. Through initial training with ICRefine we see that seven of them reach $\geq$95% accuracy after Phase 0 alone (see Appendix B) and are excluded. The five non-ceiling tasks are `geometric_shapes` (GS), `formal_fallacies` (FF), `snarks` (SN), `disambiguation_qa` (DQ), and `causal_judgement` (CJ). AGIEval stress tests [Zhong et al., 2023] use three English tasks: `lsat_ar` (LAR, $\approx$28% accuracy), `lsat_lr` (LLR, $\approx$85%), and `logiqa_en` (LQA, $\approx$54%), spanning floor, ceiling, and mid-range baseline accuracy regimes. The source model is `gpt-4.1-mini`. Target models are GPT-4.1, Gemini-2.0-flash, Llama-3.3-70B, and Claude-3.7-sonnet. The static CS-ICL baseline we are comparing against is a single-pass cheatsheet generated by GPT-4.1 from all 100 training examples, following Honda et al. [2025]. with GPT-4.1 as the CS-ICL generator and gpt-4.1-mini as the ICREFINE source model, we establish a conservative comparsion with the baseline model having a stronger generation capacity than the refined pipeline's source model. All evaluations use reasoning-first (RF) scoring: the model reasons step by step using the cheatsheet, and the response must end with `VERDICT: (X)` as its final line (parsed by scanning from the bottom of the response). We run the main pipeline and key ablations three times with independent seeds and report 3-seed means; 2–4 pp variation is expected from API nondeterminism.

5 Results: Source Gains Do Not Imply Transferable Distillation

5.1 BIG-Bench Hard: Primary Diagnostic

Source-gain/target-loss bifurcation. The clearest evidence of source-specific patching comes from Phase 2 case studies on `geometric_shapes`: the same stage adds $+5.0$ pp for the source model while reducing transfer accuracy by -3.3 pp for Claude and -6.7 pp for Llama. Source gain with target regression is the signature of distilling source-private failures ($F_s \setminus F_t$): corrections that activate on inputs the source fails but targets already answer correctly or just dont fit into the target model's reasoning pattern. It is worth noting that this sign reversal is stronger evidence than aggregate underperformance. This shows that the added artifact is not merely ineffective, but actively selects a behavior useful to the source model yet harmful to at least some targets. The contrasting case appears in Section 8: the holistic v7 achieves positive transfer of $+1.2$–$+3.8$ pp on `causal_judgement` and $+8.0$–$+23.0$ pp on `geometric_shapes` across model families, consistent with structurally shared failure modes on those tasks.

Aggregate transfer deficit. Table 3 reports average RF accuracy over the five non-ceiling tasks. The oracle-injected ICREFINE pipeline falls below CS-ICL for all four target families: GPT-4.1 (-3.4 pp), Gemini-2.0 (-1.0 pp), Llama-3.3 (-2.6 pp), and Claude-3.7 (-3.7 pp). To make sure the result is statistically significant, we perform a task-resampling bootstrap CIs confirm that the GPT-4.1 and Claude deficits exclude zero (GPT-4.1: $[-6.2, -0.7]$; Claude: $[-6.6, -0.9]$). Figure 1 shows the case-study contribution Δ_{cs} across tasks and models. We observe a task-structured pattern from this: CJ case studies regress nearly all target models (uniformly -0.8 to -4.2 pp), GS shows heterogeneous effects, and FF is near-neutral.

Table 3: Non-train transfer: average RF accuracy on five non-ceiling BBH tasks. CS-ICL: static one-shot baseline (gpt-4.1 generated). Full: oracle-injected, partition-based pipeline (v3). Δ: Full minus CS-ICL (pp). 95% CI bootstraps over five tasks; * excludes zero. **All columns are 3-seed means.**

Model	**CS-ICL**	**Full**	**Δ**	**95% CI**
GPT-4.1	87.1%	83.7%	-3.4	$[-6.2, -0.7]^*$
Gemini-2.0	81.4%	80.4%	-1.0	$[-3.5, +0.6]$
Llama-3.3	78.3%	75.7%	-2.6	$[-6.8, +1.2]$
Claude-3.7	87.7%	84.0%	-3.7	$[-6.6, -0.9]^*$

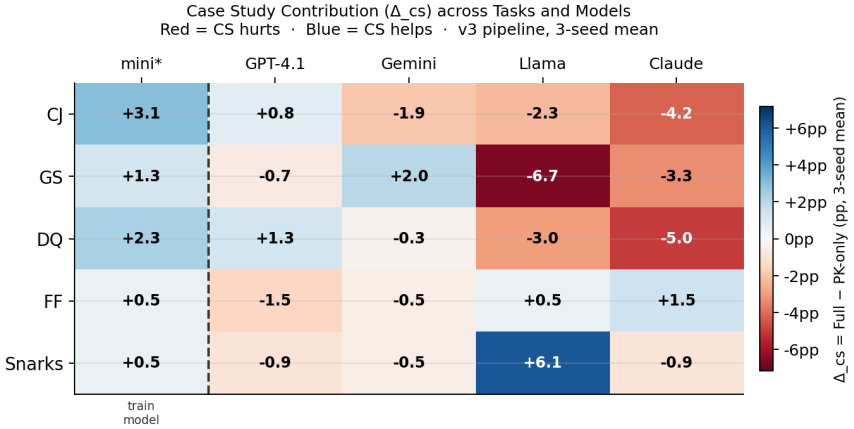

Figure 1: Case-study contribution Δ_{cs} (full minus PK-only, in pp) across 5 tasks and 5 models (ICREFINE with 3-seed means). Dashed line separates the source model (mini) from target models.

Figure 1 shows that degradation is task-structured rather than merely model-structured.

5.2 AGIEval Stress Test and Cross-Benchmark Summary

On AGIEval, ICREFINE shows a different aggregate pattern from BBH: GPT-4.1 gains $+3.7$ pp mean (largest on `logiqa_en`, $+10.6$ pp), while Gemini is near-neutral (-0.4 pp) and Llama slightly positive ($+0.7$ pp). Oracle injection amplifies within-family GPT-4.1 gains to $+16.5$ pp mean, which is $7\times$ more than cross-family targets ($+2.7$ pp Gemini, $+2.2$ pp Llama). We detailed this pattern as the family-selective oracle mechanism in Section 7 (Table 20, Appendix H).

Unlike the BBH runs, AGIEval shows family-selective transfer: refinement helps the same-family GPT-4.1 target more than cross-family targets, while BBH shows a negative-transfer regime. This supports the view that transfer depends on source-target failure alignment rather than source accuracy alone. Table 6 (Section 8) places all pipeline conditions side by side.

6 Failure Geometry: When Is Refinement Worth Attempting?

We next evaluate whether baseline failure-set Jaccard can screen for transfer risk. Table 5 summarizes five regimes observed across BBH and AGIEval. We measure the failure overlap by computing pairwise failure-set Jaccard on the CS-ICL baseline (full matrix in Table 7, Appendix C).

Five transfer-risk regimes. We separate the task sapce into five regimes based on their source-failure representativeness. Table 4 maps failure geometry to observed transfer outcome across our tasks. In our experiments, low J (< 0.3) is a reliable warning: corrections induced from source failures will likely be source-specific. High J is necessary but not sufficient: specific failure mechanisms (Section 7) can still produce source-specific artifacts even when overlap is favorable.

Table 4: Transfer-risk regimes identified across BBH and AGIEval tasks. J_{src}: Jaccard between gpt-4.1-mini and GPT-4.1 on CS-ICL baseline. Full Jaccard matrix in Appendix C.

Regime	**Task**	J_{src}	**Context**	**Transfer outcome**
High J, structured	GS (BBH)	0.86	mid-range	Task-related if PK quality high; oracle/overfitting can override
Moderate J, heterogeneous	CJ (BBH)	0.70	mid-range	Mixed; mechanism-dependent
Near-zero J	FF (BBH)	0.25	near-ceiling	Source-specific or non-transfer
Floor inflation	LAR (AGIEval)	0.66	≈28%	High J uninformative; floor-driven overlap
Ceiling contraction	LLR (AGIEval)	0.30	≈85%	Low J warns correctly; model-private residuals

These regimes serve as controls for the diagnostic. GS shows that high overlap can expose transferable structure, while FF shows that low cross-family overlap leaves little shared failure signal for source-failure-targeted rules. AGIEval calibrates the diagnostic at accuracy extremes: LAR's overlap is floor-driven, while LLR's sparse residual errors are model-private. In all cases, overlap must be interpreted together with baseline accuracy.

Jaccard screens direction; it does not predict gain. We examine this claim through different tasks. GS (high J) is the positive control in our story: EA Phase 1 improves PK-only transfer to GPT-4.1 by $+4.0$ pp (Table 9, Appendix D), suggesting task-related content is accessible when failure structure is shared. However, ICREFINE still regresses Llama (-6.7 pp), because oracle contamination and partition overfitting co-determine the outcome. FF (near-zero cross-family J, $J = 0.000$ for GPT↔Gemini) is the negative control: source-failure-targeted corrections are likely to produce source-specific patches. LSAT-AR (≈28%) has the highest J_{src} (0.66, Table 4), but overlap is floor-driven; LSAT-LR (≈85%) has $J_{\text{src}} = 0.30$ (Table 4), correctly warning of model-private residuals. Across 15 BBH task-target pairs, Jaccard is not significantly correlated with transfer magnitude for either ICREFINE or holistic v7 (Spearman $\rho = -0.39$, $p = 0.15$ and $\rho = -0.19$, $p = 0.50$; Fig. 2, Appendix C), reinforcing its role as a one-sided risk screen rather than a gain predictor.

7 Failure-Mode Identification: How Distillation Selects Source-Specific Patches

To understand transerability inefficiency under high Jaccard cases, We identify three failure mechanisms from the ICREFINE pipeline.

Identification logic. We identify a mechanism when three signals align: (i) a targeted intervention changes transfer in the predicted direction; (ii) the effect is task-structured rather than a uniform model weakness; and (iii) artifact or failure-set evidence explains why the added content activates on source-private rather than shared failures (see Table 2, §3.2). Oracle contamination satisfies this through the no-oracle CJ ablation and the AGIEval family-selective oracle gains; partition overfitting through Phase 2 source-gain/target-loss and repeated narrow case-study atoms; and bootstrap convergence through the EA Phase 1 result that eliminates the need for case studies.

Primary mechanism: oracle contamination. Oracle contamination is the most surprising failure mode we found. We use the source model to generate reasoning favoring the ground truth label for the training sets that mimics the CoT of the model when handling such case. This gold CoT is provided to the model when generating PK modification or case study with the corresponding failed cases. This gold reasoning appears to provide privileged task knowledge to the model in that it gives the distillation prompt the correct reasoning path and should help the system abstract a better generalizable rule. However, transferability failure occurs when this reasoning path is too item-specific. Instead of abstracting the task invariant, the prompt compresses the oracle trace into a source-conditioned patch: a rule that resembles the source failure's local reasoning trajectory and activates on target-correct inputs.

Table 5: Mechanisms that push distillation toward source-specific patches. **Bold** = primary/secondary role. Evidence references are 3-seed means.

Mechanism	Role	Evidence	Design response
Oracle contamination	**Primary:** gold CoT over-specifies source-family reasoning traces	CJ no-oracle improves all models $+1.9$–$+3.8$ pp; AGIEval oracle gain is $7\times$ larger for GPT-4.1 than cross-family targets	Remove oracle CoT; validate across families
Partition overfitting	**Secondary:** local memory atoms overfit recurring source-failure clusters	CJ dispersed failures; FF repeated illicit-conversion atoms; GS source gain but target split	Prefer holistic oracle-free rewrites over partition memories
Bootstrap convergence	Supporting: incomplete PK pushes residual errors into high-specificity Phase 2	GS Phase 1 accepts zero patches; EA improves PK-only $+6.0$ pp and eliminates case studies	Improve global PK search before local memory

The diagnostic prediction is asymmetric. If oracle CoT acts as task knowledge, removing it should reduce transfer or at least leave it unchanged. If it acts as source-conditioned over-specification, removing it should improve transfer, especially on heterogeneous tasks where one reasoning trace is a poor proxy for the task. From experiments we observe that removing oracle access from CJ's Phase 1 and Phase 2 improves every model by $+1.9$ to $+3.8$ pp (E3; Table 14, Appendix F), supporting the interpretation that oracle traces were actively biasing the artifact toward non-transferable reasoning patches. AGIEval shows the complementary side: oracle injection is not uniformly harmful but family-selective. GPT-4.1, the same family as the gpt-4.1-mini source, gains $+16.5$ pp mean while Gemini and Llama gain only $+2.7$ and $+2.2$ pp. This $7\times$ gap suggest that oracle traces transfer when the target shares the source family's failure structure but provide little model-independent signal. Furthermore, when J is low (`lsat_lr`, $J = 0.296$), even the within-family gain collapses ($+0.9$ pp GPT-4.1) and cross-family models are slightly harmed (Table 20, Appendix H).

Secondary mechanism: partition overfitting. Partition overfitting is the structural failure mode of ICREFINE Phase 2. The partitioner observes recurrence within F_s, but recurrence inside the source failure set is not the same as recurrence across models. When source failures are heterogeneous or source-private, Phase 2 promotes narrow clusters into ACTIVATE IF/WHY atoms that target $F_s \setminus F_t$. This differs from oracle contamination: the case study need not be generated from a gold reasoning trace to be harmful. The problem is that the partitioning criterion treats source-local recurrence as task-level structure. On CJ, the 35 training failures span six heterogeneous causal sub-types. We observe that no partition yields a broad high-fix rule without regressing others, and Phase 2 reduces transfer for Claude (-4.2 pp), Gemini (-1.9 pp), and Llama (-2.3 pp). On FF, three of four case studies target illicit conversion of conditionals, showing greedy fixation on one local pattern while leaving other fallacy types uncovered. On GS, concentrated SVG-parsing failures let Phase 2 improve the source model by $+5.0$ pp, but target effects split by family, revealing that even concentrated source clusters transfer only when targets share that failure mode.

Supporting mechanism: bootstrap convergence. Even though bootstrap convergence is not the central source of source-specific content, it explains why the pipeline enters a high-specificity Phase 2 regime. When Phase 1 stops with a gate-passing but incomplete PK, residual errors are delegated to Phase 2, whose representation is local by construction. On GS, standard Phase 1 applies zero patches because the bootstrap PK satisfies the gate despite leaving accuracy on the table. EA Phase 1 improves PK-only accuracy from 72.7% to 78.7% ($+6.0$ pp, 3-seed mean) and generates zero accepted case studies, supporting the view that better global task compression removes the demand for local source-failure memory.

Synthesis. These transferability failure mechanisms differ in where specificity enters: oracle contamination over-specifies the reasoning trace, partition overfitting over-specifies the failure cluster, and bootstrap convergence under-specifies the global rule so that local patches become necessary. Together, they explain how a refinement pipeline can add information while reducing transfer and point to distinct design responses for each pressure source.

Table 6: Cross-benchmark summary—mean Δ vs. CS-ICL (pp) for all pipeline conditions and model families. ICREFINE: oracle-injected v3, 3-seed means. Oracle: Experiment D (3 AGIEval tasks $\times$ 3 seeds). Holistic v7: single seed, gpt-4.1-mini CS-ICL warmstart. ProTeGi: beam-3, 4 rounds, best-by-train. GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b. † Single seed; multi-seed results planned for a future version.

Condition	**Bench**	**ΔGPT**	**ΔGem.**	**ΔLlm.**	**Trend**
ICREFINE	BBH (5 tasks)	-3.4	-1.0	-2.6	All families negative
ICREFINE	AGIEval (3 tasks)	$+3.7$	-0.4	$+0.7$	Intra-family positive
Oracle ICREFINE	AGIEval	$+16.5$	$+2.7$	$+2.2$	$7\times$ family gap
Holistic v7†	BBH (3 tasks)	$\mathbf{+3.7}$	$\mathbf{+5.8}$	$\mathbf{+8.7}$	GS dominates; all positive
Holistic v7†	AGIEval (2 tasks)	$\mathbf{+1.2}$	$\mathbf{+3.7}$	$\mathbf{+6.1}$	All positive
ProTeGi	BBH (6 tasks)	$+0.2$	$+0.6$	-1.3	Near-neutral
ProTeGi	AGIEval (3 tasks)	$+3.3$	-0.5	$+0.7$	Intra-family positive

8 Conservative Controls and External Checks

Holistic v7. *Note: v7 results are from a single seed; multi-seed runs are planned for a future version of this paper.* Holistic v7 addresses all three specificity pressures simultaneously by removing oracle injection and partition-based case studies from the loop. Starting from a gpt-4.1-mini CS-ICL warmstart, the pipeline iterates holistic whole-cheatsheet rewrites validated by fix-rate and regression gates, with rollback to the best-so-far cheatsheet and early stopping after three consecutive non-improving iterations.

Evaluated on five tasks (CJ, DQ, GS from BBH; LAR, LQA from AGIEval) with non-train models GPT-4.1, Gemini-2.0-flash, and Llama-3.3-70b, the pipeline achieves positive transfer on `geometric_shapes`: $+8.0$ pp (GPT-4.1), $+16.0$ pp (Gemini), $+23.0$ pp (Llama). On `causal_judgement` the pattern is also positive across all families: $+3.8$ pp (GPT-4.1), $+2.7$ pp (Gemini), $+1.2$ pp (Llama). `disambiguation_qa` is mixed (-0.7 pp GPT, -1.3 pp Gemini, $+2.0$ pp Llama), consistent with its moderate Jaccard. On AGIEval, `logiqa_en` gains $+2.4$–$+3.5$ pp across families; `lsat_ar` gains $+4.9$ pp (Gemini) and $+8.7$ pp (Llama), though GPT-4.1 shows no gain over our gpt-4.1-mini CS-ICL baseline, consistent with the floor-driven Jaccard inflation on this task. Full per-task results are in Table 15 (Appendix F).

Table 6 places holistic v7 alongside ICREFINE and ProTeGi. The contrast is sharp: ICREFINE is negative across all BBH families (-3.4, -1.0, -2.6 pp), while holistic v7 is positive on both BBH ($+3.7$, $+5.8$, $+8.7$ pp mean) and AGIEval ($+1.2$, $+3.7$, $+6.1$ pp mean). ProTeGi, which also avoids oracle injection and partition memories, shows near-neutral BBH aggregate ($+0.2$, $+0.6$, -1.3 pp), sitting between ICREFINE's regressions and holistic v7's gains. Both holistic v7 and ProTeGi avoid the large negative swings of ICREFINE, consistent with the view that removing oracle injection and partition-based case studies is the key design choice for preserving transferability.

Artifact contrast. On `geometric_shapes`, the holistic rewrite consolidates SVG geometry parsing heuristics into a coherent prior-knowledge block that targets the shared structural failure mode across models. This is task-structural: the revision encodes the geometric reasoning the task requires, not traces specific to one source model. By contrast, Phase 2 case studies on `causal_judgement` encode narrow causal subtypes such as joint-AND causation or redundant contributors that do not cover the task's broader failure modes. The same study can activate on inputs target models already handle correctly and cause regressions. This contrast illustrates the selection problem directly: holistic rewrites conditioned on the full failure set are more likely to target $F_s \cap F_t$, while partition-based case studies risk encoding $F_s \setminus F_t$.

ProTeGi provides an optimizer-side stress test. We address the possibility that the negative-transfer pattern is an artifact of ICREFINE's partition-based design by including another established refinement mechanism. We compare with ProTeGi [Pryzant et al., 2023] under the same cross-model criterion (full results in Tables 16–17, Appendix G). As Table 6 shows, ProTeGi's BBH aggregate ($+0.2$, $+0.6$, -1.3 pp) is near-neutral and avoids ICREFINE's large regressions, but falls short of holistic v7's consistent positive gains. The task-structured pattern follows failure geometry:

`date_understanding` yields the largest Gemini gain ($+5.3$ pp), consistent with shared arithmetic failure structure, while `disambiguation_qa` produces the sharpest Gemini regression (-4.7 pp), consistent with its low source–Gemini Jaccard ($J = 0.333$, Table 7). On AGIEval, ProTeGi shows the same intra-family positive / cross-family neutral pattern as ICREFINE: GPT-4.1 gains $+3.3$ pp mean while Gemini (-0.5 pp) and Llama ($+0.7$ pp) are near-zero. Taken together with holistic v7, the pattern across all three systems is consistent: removing oracle injection and partition-based memories (holistic v7) achieves the largest positive transfer; using a different gradient-based optimizer without those features (ProTeGi) reaches near-neutral; keeping oracle injection and partition memories (ICREFINE) causes regression. The loss-of-transferability risk thus appears to be in failure-targeted distillation with oracle injection and local partitioning, not in failure-targeted distillation itself.

9 Discussion

Our results suggest that failure-driven textual distillation is a selection problem: the pipeline must select between task-level structure and source-specific patches. Yet, Source accuracy alone cannot distinguish these outcomes, which means we need to adopt cross-model transfer and baseline failure overlap as necessary diagnostics. The strongest specificity pressures come from oracle CoT and partition-local memory, while oracle-free holistic rewriting provides a conservative alternative when shared failure structure exists.

Limitations

Most of our test sets are limited in size(87–250 items per BBH task) so the conclusion may not easily generalize to refinement training on much larger datasets. The three failure mechanisms we identified are diagnostic rather than exhaustive, and Jaccard is a necessary-condition screen evaluated retrospectively on our tasks rather than as a validated deployment policy. It is also worth noting that ProTeGi is included as an optimizer stress test, not a fully tuned benchmark, and the holistic v7 positive results are single-seeded. Multi-seed means for v7 are planned for a future version of this paper to establish robustness. Cross-model transfer is operational evidence instead of proof of task-related content, which means surviving improvements may still reflect shared LLM biases rather than task-structural invariants.s

Related work

CS-ICL [Honda et al., 2025] is our direct baseline: it generates a cheatsheet from failure-identified demonstrations in a single LLM pass. We do iterate refinement based on this process and ask whether the accumulated content behaves like task knowledge or a source-specific patch. Prompt-optimization methods [Zhou et al., 2023, Yang et al., 2024, Pryzant et al., 2023, Khattab et al., 2024, Yuksekgonul et al., 2024, Fernando et al., 2024] treat the prompt as an optimization target scored on source-model performance. Unlike that line of work, we ask what the optimized artifact distills when transferred to a different model family, and show the answer depends on failure-set overlap measurable before refinement begins. Self-Refine [Madaan et al., 2024] and Reflexion [Shinn et al., 2024] iterate on per-instance outputs using self-generated feedback. our pipeline targets a shared cheatsheet artifact, so source-specificity accumulates across iterations rather than being reset at inference. Wang et al. [2025] characterizes prompt degradation across model versions as distribution shift. We examine the complementary case in which prompts degrade by construction because failure-targeted refinement may distill source-private failure modes into the shared artifact.

Conclusion

Textual distillation is not only a compression problem: it is a selection problem over what kind of knowledge is compressed. Failure-set Jaccard gives a cheap check, before any refinement runs, on whether the source model's mistakes are likely to be the right training signal, but transferability failure can still occur at high J due to oracle contamination, partition overfitting, or bootstrap convergence. Conservative oracle-free, validation-gated holistic refinement avoids most regressions and handles the three mechanisms we identify.

References

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-referential self-improvement via prompt evolution. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Ukyo Honda, Soichiro Murakami, and Peinan Zhang. Distilling many-shot in-context learning into a cheat sheet. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. URL `https://arxiv.org/abs/2509.20820`.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Hira, Ashutosh Zhong, Tom Kwiatkowski, Christopher Potts, Percy Liang, and Matei Zaharia. DSPy: Compiling declarative language model calls into self-improving pipelines. In *The Twelfth International Conference on Learning Representations*, 2024.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skye Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Sean Welleck, Peter Clark, Antoine Bosselut, Peter Clark, Sameer Singh, and Ari Holtzman. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chenguang Zhu, and Michael Zeng. Automatic prompt optimization with "gradient descent" and beam search. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7957–7968, 2023.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging BIG-bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*, 2022.

Yaxuan Wang, Quan Liu, Zhenting Wang, Zichao Li, Wei Wei, Yang Liu, and Yujia Bao. PromptBridge: Cross-model prompt transfer for large language models. *arXiv preprint arXiv:2512.01420*, 2025.

Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *The Twelfth International Conference on Learning Representations*, 2024.

Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Zhi Huang, Carlos Guestrin, and James Zou. TextGrad: Automatic "differentiation" via text. *arXiv preprint arXiv:2406.07496*, 2024.

Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yining Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. AGIEval: A human-centric benchmark for evaluating foundation models. *arXiv preprint arXiv:2304.06364*, 2023.

Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. Large language models are human-level prompt engineers. In *The Eleventh International Conference on Learning Representations*, 2023. URL `https://openreview.net/forum?id=92gvk82DE-`.

A Full ICREFINE Algorithm

A.1 Phase 0: Bootstrap

The pipeline initialises the PK by prompting the train model with the first $n_{\text{boot}} = 75$ training examples and requesting a concise rules-and-heuristics summary (single pass, no iteration).

A.2 Phase 1: Prior-Knowledge Patching

At each iteration: (1) Score all 100 training items; collect failures $\mathcal{F}_t$. (2) Sample $\min(15, |\mathcal{F}_t|)$ failures; optionally inject gold CoT (`item["reason"]`) as a "Correct reasoning" contrast block. (3) Generate one candidate PK revision (temperature 0.3). (4) Apply fix-rate gate (≥ 0.20) and regression gate (≤ 0.20) in sequence. (5) If both pass, replace PK and re-score. Phase 1 exits when training accuracy ≥ 0.85, two consecutive iterations produce no accepted patch, or 4 iterations are exhausted.

A.3 Phase 2: Partition-Based Case Study Generation

Failures are partitioned by structural error type (deterministic, no LLM). Only partitions with ≥ 3 failures are processed. For each partition: generate $N = 3$ candidate case studies (temperatures 0.3, 0.6, 0.9); select the highest fix-rate candidate; apply fix-rate gate (≥ 0.30) and regression gate (≤ 0.15). Phase 2 exits when all partitions retire, two consecutive outer iterations add no case study, or 5 outer iterations are exhausted. A similarity gate checks for duplicate or mergeable case studies when ≥ 3 exist.

A.4 Evolutionary Algorithm Phase 1

Three members are maintained: *conservative* (temperature 0.2), *generative* (temperature 0.7), and *structural* (temperature 0.3). All are scored on a fixed 80%/20% train/validation split. Candidates are accepted if they satisfy: (a) size $\leq 12{,}000$ characters; (b) curriculum net-gain $n_{\text{fixed}} \geq \lambda(g) \cdot n_{\text{regressed}}$, where $\lambda(g)$ ramps from 1.0 (generation 1) to 2.0 (final generation); (c) absolute regression cap $n_{\text{regressed}} \leq \max(3,\, 0.20 \cdot |\mathcal{C}_{\text{pool}}|)$. After each generation, the top $k = 2$ members by validation accuracy survive and are recombined into a child PK via crossover (temperature 0.3). The algorithm exits after 4 generations or when all members are static for ≥ 2 consecutive generations.

A.5 Oracle Injection

Gold CoT (`item["reason"]`) is injected into both Phase 1 and Phase 2 failure-analysis prompts as a "Correct reasoning" contrast block. Priority order: task-specific pre-computed oracle > `item["reason"]` dataset field > external oracle CSV lookup. Disabled in the E3 condition (`-no-oracle`) and in the holistic v7.

A.6 Transfer Threshold Derivation

Let $F_M(\mathcal{C}) = \{x : h_M(x; \mathcal{C}) \neq y(x)\}$ be the failure region and $R_M(\mathcal{C}) = \{x : h_M(x; \mathcal{C}) = y(x)\}$ the correct region. The marginal gain of adding a knowledge atom c to $\mathcal{C}$ for model M is

$$G_M(c \mid \mathcal{C}) = \Pr\big[x \in F_M,\ h_M(x; \mathcal{C} \cup \{c\}) = y(x)\big] - \Pr\big[x \in R_M,\ h_M(x; \mathcal{C} \cup \{c\}) \neq y(x)\big].$$

With activation mass $p_M(c) = \Pr[a_c(x) = 1]$, activation precision $\pi_M(c) = \Pr[x \in F_M \mid a_c(x) = 1]$, fix rate $f_M(c)$, and regression rate $r_M(c)$:

$$G_M(c \mid \mathcal{C}) = p_M(c)\big[\pi_M(c) f_M(c) - (1 - \pi_M(c)) r_M(c)\big].$$

Rearranging gives Eq. (1): $\pi_M(c) > r_M(c)/(f_M(c) + r_M(c))$. For a source-failure-targeted correction with $A_c \subseteq F_s$, $\pi_{M_{\text{tgt}}}(c) \leq |F_s \cap F_t|/|A_c|$; when $A_c \approx F_s$, this approaches $\rho_{s \to t} = |F_s \cap F_t|/|F_s|$, linking Jaccard screening to the transfer threshold.

B Additional Experimental Details

Reasoning-first (RF) scoring. All conditions use RF scoring. The scoring prompt instructs the model to work through the problem step by step using the cheatsheet, then end its response with the verdict as the final line: `VERDICT: (X)` (no other text after it). The verdict is extracted by scanning the response from the bottom and taking the last line that begins with `VERDICT:`. Verdict-first CoT produces a severe format artifact (web_of_lies: 42–66% verdict-first vs. 99–100% RF); all comparisons are internally consistent under RF.

CS-ICL baseline. Generated by gpt-4.1 (not gpt-4.1-mini) from all 100 training examples, matching Honda et al. [2025]; contains no case studies and undergoes no iterative refinement.

Task list. We evaluated 11 BBH tasks. Seven were excluded from detailed ICREFINE analysis because the Phase 0 source model reached ≥95% training accuracy: `boolean_expressions`, `navigate`, `web_of_lies`, `logical_deduction_three`, `sports_understanding`, and the source-model-ceiling variants of `formal_fallacies` and `snarks`. The remaining five non-ceiling tasks are `geometric_shapes`, `formal_fallacies` (non-source-ceiling), `snarks` (non-source-ceiling), `disambiguation_qa`, and `causal_judgement`. We retain `formal_fallacies` and `snarks` in the five-task transfer analyses because non-source evaluation models leave measurable error on these tasks; we note this distinction explicitly where it matters for interpretation.

C Full Jaccard Matrix and Transfer Scatter

Table 7 reports the complete pairwise failure-set Jaccard matrix for all eight tasks (5 BBH + 3 AGIEval) and all six model pairs on the CS-ICL baseline. Figure 2 plots failure-set Jaccard vs. transfer Δ for 15 task–model pairs, confirming no significant monotonic trend consistent with Jaccard as a one-sided screen.

Table 7: Pairwise failure-set Jaccard overlap $J(M_1, M_2; T)$ on CS-ICL baseline (one seed per task/model; failure sets on test split). High J = structurally shared failures; low J = model-specific failures. SN has few failures (≤4 per model); Jaccard is noisy. AGIEval rows show source-model accuracy to contextualise floor/ceiling effects.

Task	mn↔GPT	mn↔Gem	mn↔Llm	GPT↔Gem	GPT↔Llm	Gem↔Llm
BBH tasks						
CJ	0.704	0.471	0.471	0.533	0.438	0.515
DQ	0.400	0.333	0.296	0.286	0.458	0.229
FF	0.250	0.133	0.227	0.000	0.136	0.231
GS	**0.864**	0.545	0.559	0.543	0.514	0.467
SN	0.500	0.500	0.333	1.000	0.500	0.500
AGIEval tasks (source-model accuracy)						
LAR (28%)	0.660	0.716	0.695	0.680	0.660	0.646
LLR (85%)	0.296	0.333	0.364	0.333	0.396	0.278
LQA (53%)	0.624	0.506	0.595	0.528	0.549	0.535

LAR = LSAT-AR; LLR = LSAT-LR; LQA = LogiQA-EN. mn = gpt-4.1-mini.

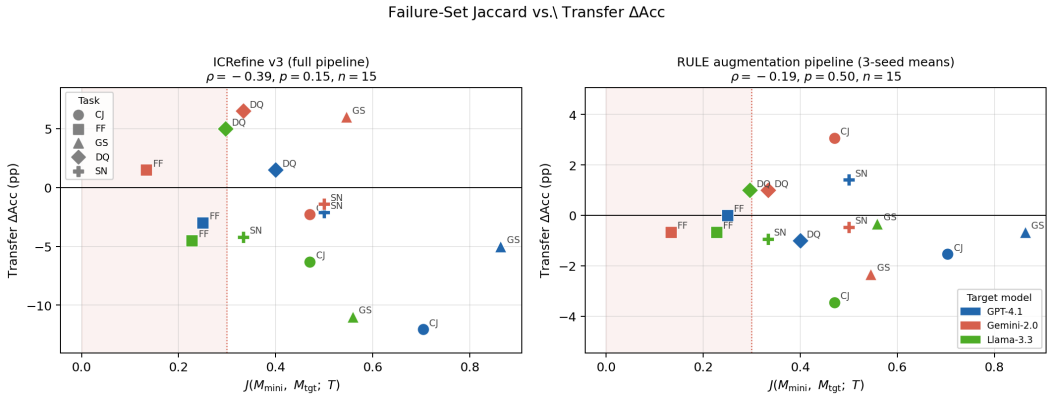

Figure 2: Failure-set Jaccard $J(M_{\text{mini}}, M_{\text{tgt}}; T)$ vs. transfer ΔAcc (pp) for 15 task–target-model pairs. *Left*: ICREFINE v3 full pipeline (Spearman $\rho = -0.39$, $p = 0.15$). *Right*: holistic v7 ($\rho = -0.19$, $p = 0.50$). Neither panel shows a significant monotonic trend, consistent with Jaccard as a one-sided screening diagnostic. Task shapes: **CJ**= ○, **FF**= □, **GS**= △, **DQ**= ◇, **SN**= +.

D Additional Tables and Per-Seed Results

D.1 Phase Contribution Breakdown

Figure 3 shows RF test accuracy by pipeline phase across all 11 BBH tasks for the train model (gpt-4.1-mini), providing context for the ceiling-task exclusion criterion used in Section 4.

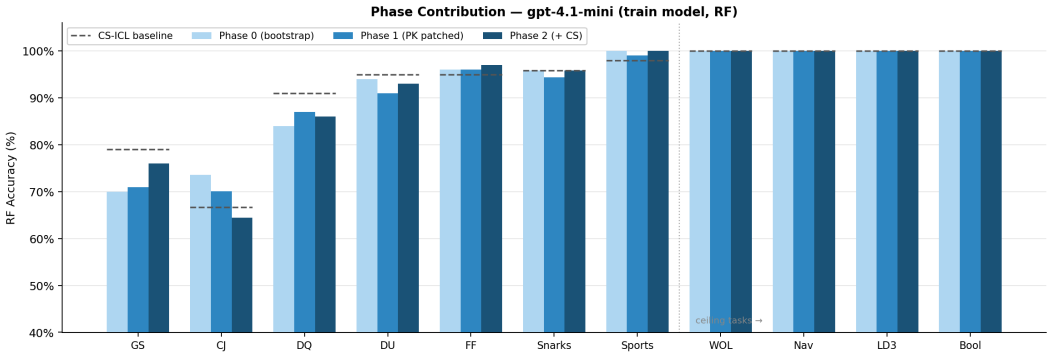

Figure 3: RF test accuracy by pipeline phase for gpt-4.1-mini across all 11 BBH tasks. Seven tasks reach $\geq$95% after Phase 0 alone. Phase 2 produces the largest task-level shifts on GS ($+5.0$ pp) and CJ (-5.7 pp).

D.2 Seed Variance Details

Figure 4 and Table 8 give the full per-seed breakdown for the two ablations most central to the main claims: GS PK-only (EA vs. v3) and CJ full (E3 vs. v3).

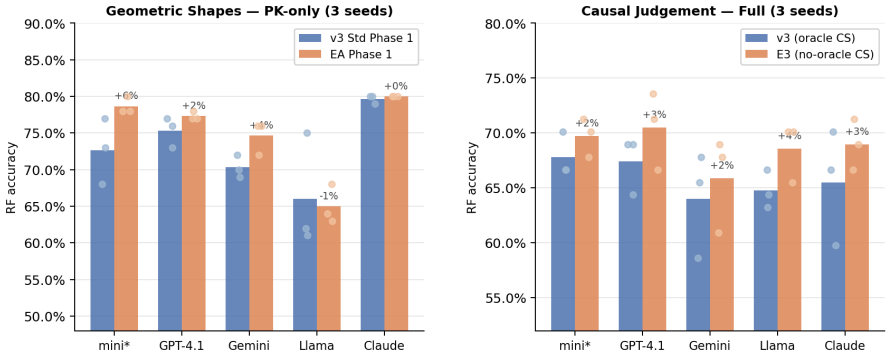

Figure 6: Per-seed (dots) and 3-seed mean (bars) RF accuracy for the two key ablation comparisons.
Left: EA Phase 1 raises GS PK-only by +2–6 pp across all models; EA results are more stable (Llama σ=4.7% vs v3 σ=10.4%).
Right: Removing oracle access (E3) improves CJ full accuracy for all five models (+1.9–3.8 pp), consistent with oracle contamination as a contributing factor; heterogeneous failure distribution is a plausible co-contributor (see Appendix B).

Figure 4: Per-seed (dots) and 3-seed mean (bars) RF accuracy for two key ablation pairs. *Left*: EA Phase 1 raises GS PK-only accuracy by $+2$–$+6$ pp across models; EA seed-to-seed variance is lower (Llama $\sigma = 4.7\%$ vs. 10.4%). *Right*: Removing oracle access (E3) improves CJ full accuracy for all five models by $+1.9$–$+3.8$ pp, consistent with oracle contamination.

D.3 EA vs Standard Phase 1

Table 9 reports per-task train-model accuracy for EA vs. standard Phase 1, isolating Phase 1 quality from Phase 2 effects by using PK-only cheatsheets.

D.4 Shared-Failure Filtering Results

Table 10 reports the full per-task results for the failure-intersection training experiment (Section 7).

Table 8: Per-seed RF accuracy for two key comparisons (GS PK-only; CJ full). v3: standard Phase 1. EA: evolutionary Phase 1. E3: oracle-free (both phases).

Task / Condition	Model	S1	S2	S3	Mean	σ
GS PK-only (v3)	mini	71	72	75	72.7	2.1
	GPT-4.1	80	83	82	81.7	1.5
	Claude	84	85	90	86.3	3.2
	Gemini	77	79	81	79.0	2.0
	Llama	51	56	71	59.3	10.4
GS PK-only (EA)	mini	79	79	78	78.7	0.6
	GPT-4.1	85	86	86	85.7	0.6
	Claude	89	88	91	89.3	1.5
	Gemini	82	82	84	82.7	1.2
	Llama	68	72	72	70.7	2.3
CJ full (v3)	mini	64	72	68	68.0	4.0
	GPT-4.1	60	71	71	67.3	6.4
	Claude	66	65	65	65.3	0.6
	Gemini	62	64	66	64.0	2.0
	Llama	61	65	68	64.7	3.5
CJ full (E3, no oracle)	mini	69	71	69	69.7	1.2
	GPT-4.1	68	71	72	70.3	2.1
	Claude	69	69	69	69.0	0.0
	Gemini	65	66	66	65.7	0.6
	Llama	66	70	70	68.7	2.3

Table 9: EA Phase 1 vs. standard Phase 1—RF test accuracy on gpt-4.1-mini. *PK only*: cheatsheet without case studies. *Full*: complete pipeline (PK + case studies). Δ PK: EA PK minus Std PK. **PK columns are 3-seed means**; CS counts and patch counts are from seed 1.

Task	Std PK	Std CS	EA PK	EA CS	EA patches	Δ PK
CJ	64.8%	1	**70.1%**	4	0	$+5.4$
GS	72.7%	2	**78.7%**	0	3	$+6.0$
Snarks	96.7%	2	95.8%	1	0	-0.9
DQ	84.7%	0	84.7%	3	1	0.0

D.5 EA+no-oracle Combined Pipeline

Table 11 reports the combined EA+no-oracle pipeline, included for completeness. This pipeline addresses two failure modes simultaneously but remains below CS-ICL, suggesting the bottleneck is structural failure-set misalignment rather than any single addressable design flaw.

D.6 Best-Combo Per-Task Oracle Upper Bound

Best-combo selects the best available intervention per task independently (EA PK-only for GS; oracle-free E3 for CJ; v3 full for FF, DQ, SN). It is not a deployable pipeline; it serves as an upper bound. The per-task oracle margin over EA+no-oracle is 0.1–2.9 pp.

D.7 Second Training Model

With gemini-2.0-flash as train model, non-train Δ vs. CS-ICL (-3.7, -2.9, -3.5 pp for GPT-4.1, Claude, Llama) are within 1 pp of the mini-trained equivalents. The task-structured pattern replicates: CJ case studies are negative regardless of train model; GS case studies benefit models sharing the train model's SVG parsing failures.

Table 10: Failure-intersection training: 3-seed mean transfer accuracy (%) on Gemini-2.0-flash and Llama-3.3-70b across three tasks. Best-by-train cheatsheet selected per condition. Bold = best per task×model. Standard training on all source failures is the baseline; intersection filtering does not improve the mean.

Model	Task	Standard	Intersect	Anti-int	Random
Gemini-2.0-flash	CJ	**67.8**	67.0	67.0	67.0
	FF	**90.3**	87.3	90.0	87.5
	DQ	79.3	**81.0**	78.7	79.7
	Mean	**79.2**	78.5	78.6	78.1
Llama-3.3-70b	CJ	67.8	**69.0**	67.4	66.3
	FF	**81.7**	80.3	79.0	81.5
	DQ	79.3	**80.3**	**80.3**	78.7
	Mean	76.3	**76.5**	75.6	75.5

Table 11: EA+no-oracle vs. CS-ICL—non-train 5-task RF average (3-seed means). Δ = EA+no-oracle − CS-ICL; 95% CI bootstraps over 5 tasks. * CI excludes zero.

Model	CS-ICL	EA+no-oracle	Δ	95% CI
GPT-4.1	87.1%	83.8%	-3.3	$[-6.3, -1.6]^*$
Gemini-2.0	81.4%	80.8%	-0.6	$[-2.7, +1.6]$
Llama-3.3	78.3%	74.7%	-3.6	$[-6.4, +0.8]$
Claude-3.7	87.7%	85.3%	-2.4	$[-5.4, -0.0]^*$

D.8 Size Ablations (Exploratory, Single-Run)

These single-run ablations on gpt-4.1-mini examine sensitivity to PK character limit (Figure 5) and Phase 2 case study count (Figure 6). All differences fall within or near the 2–4 pp evaluation noise floor.

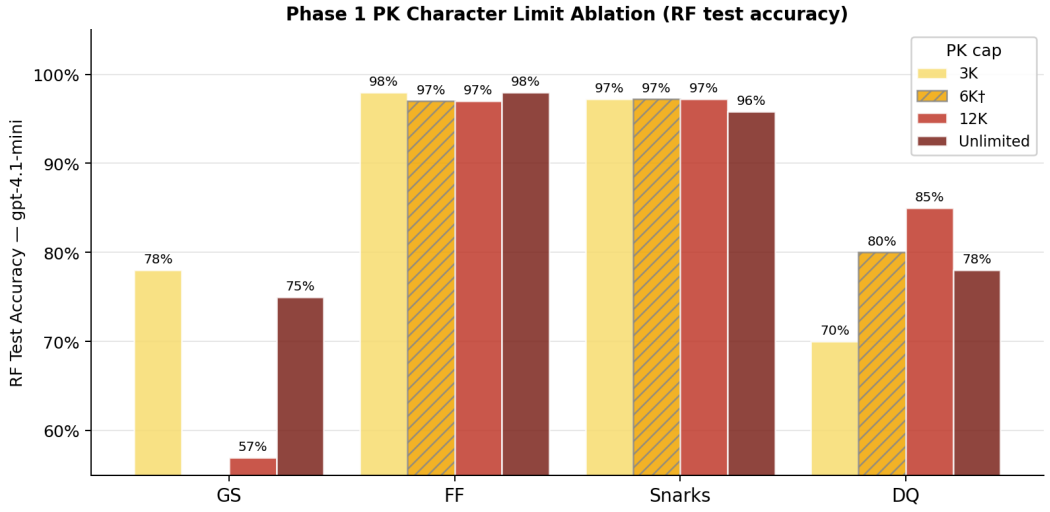

Figure 5: Phase 1 PK character limit ablation (single-run, gpt-4.1-mini only). Ordering is non-monotone across all four tasks; no cap consistently outperforms. †6K GS = 18% (anomalous, likely corrupted cheatsheet; excluded).

Table 12: Best-combo (per-task oracle) vs. EA+no-oracle vs. CS-ICL—non-train 5-task RF average (3-seed means). †Best-combo is a per-task oracle, not a deployable pipeline.

Model	CS-ICL	EA+no-oracle	Best-combo†	Δ BC vs EA
GPT-4.1	87.1%	83.8%	84.9%	+1.1
Gemini-2.0	81.4%	80.8%	81.2%	+0.4
Llama-3.3	78.3%	74.7%	77.6%	+2.9
Claude-3.7	87.7%	85.3%	85.4%	+0.1

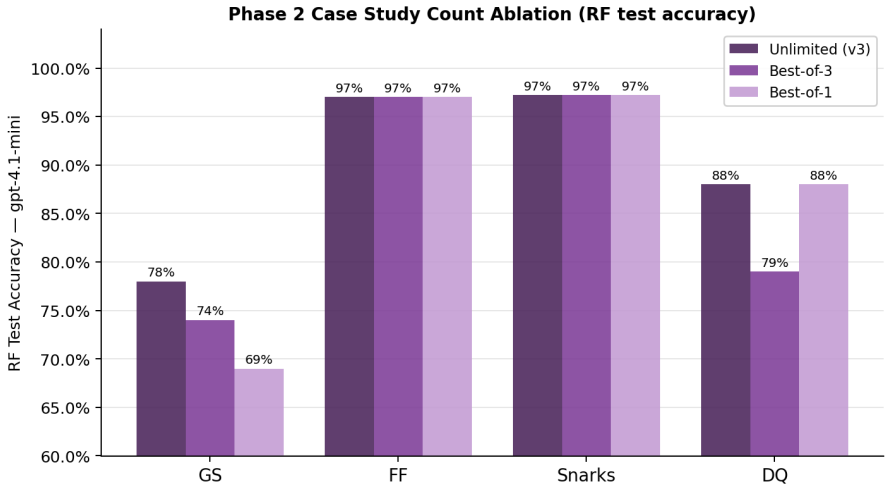

Figure 6: Phase 2 case study count ablation (single-run). GS test ordering reverses from train (unlimited wins at test; best-of-3 led in training). All other differences are within noise.

E Causal Judgement Case Study Inventory

Table 13 lists all case studies generated across seeds 1–3 of the standard oracle-enabled v3 pipeline. Seeds 2 and 3 produce structurally formatted ACTIVATE IF case studies (not oracle-contaminated), yet both seeds still reduce accuracy—implicating failure-distribution heterogeneity as a co-cause.

F Validation Gate, Augmentation Ablations, and Oracle Ablation

Validation gate contribution. For DQ, where the gate blocked all three seeds, removing the gate would have hurt Gemini by -4.7 pp mean (worst seed: -13.0 pp) while GPT and Llama would have seen only $+0.7$ pp. For GS, no-gate stage1 hurts Gemini by -2.7 pp mean while Llama gains $+2.0$ pp, showing the model-selective harm the gate correctly blocked. On tasks the gate accepted (CJ, DU, FF, SN), removing the gate changes the 4-task mean only marginally ($+0.2$ pp GPT, $+1.2$ pp Gemini, $+0.1$ pp Llama).

Oracle injection ablation (augmentation pipeline). Running the same augmentation pipeline with oracle context injected yields 4-task means of $+0.5$ pp (GPT), $+0.1$ pp (Gemini), -0.2 pp (Llama)—similar in aggregate to oracle-free ($+0.3$, $+0.7$, -0.5 pp) but with higher within-task variance. DU Gemini gains $+5.0$ pp with oracle injection versus $+1.0$ pp oracle-free; Snarks Gemini loses -6.1 pp with oracle injection versus -0.5 pp oracle-free. The oracle-free design is more stable: it avoids the Snarks-Gemini collapse while giving up part of the DU-Gemini gain.

Cross-model validation gate (diagnostic). As a post-hoc diagnostic, we score (stage1, stage2) pairs for six runs using two proxy models (Gemini-2.0-flash and Llama-3.3-70b) and keep stage2 only if it does not regress on all proxies. All six runs reject stage2. CJ and SN/seed3000 rejections are vindicated: stage2 regresses on most eval models. Two runs are over-rejected (SN/seed1000 and

Table 13: CJ case studies across three v3 pipeline seeds. Seed 1: oracle-contaminated (free-form narrative paraphrasing a training item). Seeds 2–3: structured ACTIVATE IF case studies, not oracle-contaminated. All three seeds produce case studies that reduce transfer accuracy vs. CS-ICL.

Seed	Case study title / pattern	Observation
1	*Joint causes (logical AND): garden fertilizers*	Oracle-contaminated: free-form narrative paraphrasing one training item's gold chain-of-thought; encodes one idiosyncratic joint-AND scenario.
2	*Overlapping Contributors with Non-Intersecting Effects*	Three structured ACTIVATE IF case studies addressing joint-AND / combined-harm patterns. No oracle paraphrase; conditions abstract. Covers only one causal sub-type; misses norm violations, timing, intentionality.
2	*Joint Necessary Actions Without Individual Effect*	
2	*Joint Harm Only from Combined Actions*	
3	*Redundant Contributors in Joint Outcomes*	Four structured case studies spanning overdetermination and compliance. More diverse than seed 2 but still omits norm violations and prevention chains. Two redundancy case studies are near-duplicates, paralleling FF's greedy-search fixation.
3	*Overdetermination with Redundant Contributors*	
3	*Causation by Maintaining a Sufficient Condition*	
3	*Causation by Complying with Essential Instructions*	

FF/seed3000, where stage2 is net-positive on 3 of 4 eval models), showing the gate is conservative when proxy models disagree.

CJ Oracle Ablation (E3). Table 14 reports the full per-model results for the oracle ablation on causal_judgement. E3 removes gold CoT from both Phase 1 and Phase 2 failure-analysis prompts. The consistent improvement across all five models and three seeds is the clearest causal evidence that oracle injection increases source-specific patching on CJ. Oracle removal does not fully close the gap—E3 remains 3.9 pp below CS-ICL—confirming CJ's heterogeneous failure structure is a co-cause independent of oracle access.

Table 14: Oracle ablation on causal_judgement (RF, 5 models). E3: Phase 1 and Phase 2 oracle disabled. v3 full: standard oracle-enabled pipeline. $\Delta = \text{E3} - \text{v3}$. Both columns are 3-seed means. * training model.

Model	E3 (no oracle)	v3 full	Δ
gpt-4.1-mini*	69.7%	67.8%	+1.9
GPT-4.1	70.5%	67.4%	+3.1
Claude-3.7	69.0%	65.5%	+3.5
Gemini-2.0	65.9%	64.0%	+1.9
Llama-3.3	68.6%	64.8%	+3.8

Holistic v7 per-task results. Table 15 gives the full per-task RF accuracy and Δ vs. CS-ICL for the holistic v7. All results are single-seeded; multi-seed means are planned for a future version. For AGIEval tasks, the CS-ICL baseline for GPT-4.1 is our re-scored gpt-4.1-mini-generated cheatsheet (lsat_ar: 70.4%; logiqa_en: 71.5%); Gemini and Llama baselines follow the paper's reported values.

Table 15: Holistic v7 RF accuracy and Δ vs. CS-ICL (pp)—single seed. BBH CS-ICL: gpt-4.1-generated (from main paper). AGIEval CS-ICL GPT-4.1: re-scored gpt-4.1-mini cheatsheet. AGIEval CS-ICL Gem./Llm.: paper-reported values. Last row: full ICREFINE 5-task BBH mean (3-seed, for reference). † Single seed; multi-seed results planned for a future version.

	CS-ICL (%)			Holistic v7 (%)			Δ† (pp)		
Task	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**
BBH tasks									
CJ	68.6	67.4	67.8	72.4	70.1	69.0	**+3.8**	**+2.7**	**+1.2**
DQ	84.7	79.3	81.0	84.0	78.0	83.0	−0.7	−1.3	**+2.0**
GS	79.0	68.0	63.0	87.0	84.0	86.0	**+8.0**	**+16.0**	**+23.0**
Mean (BBH 3-task)							**+3.7**	**+5.8**	**+8.7**
AGIEval tasks									
lsat_ar	70.4	31.6	29.6	70.4	36.5	38.3	0.0	**+4.9**	**+8.7**
logiqa_en	71.5	63.5	60.0	73.9	66.0	63.5	**+2.4**	**+2.5**	**+3.5**
Mean (AGIEval 2-task)							**+1.2**	**+3.7**	**+6.1**
ICREFINE full (5-task BBH)							−3.4	−1.0	−2.6

GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b.
Δ = holistic v7 − CS-ICL (pp). Source model: gpt-4.1-mini.

G ProTeGi Cross-Model Transfer Comparison

Tables 16 and 17 report the full per-task ProTeGi results under the same cross-model transfer criterion used throughout the paper. ProTeGi [Pryzant et al., 2023] is run with RF scoring, beam size 3, 4 gradient rounds, gpt-4.1-mini as the training model, and a CS-ICL cheatsheet warm-start. Prompt selection uses the round with the highest training-set score (beam leader) across rounds 1+, matching the validation-based selection used for ICREFINE. Results are 3-seed means.

Table 16: ProTeGi vs. CS-ICL baseline—3-seed mean RF accuracy on six BBH tasks. Δ = ProTeGi − CS-ICL (pp). Prompt selected by best training-set score (not test accuracy). `date_understanding` is the strongest Gemini win (+5.3 pp); `disambiguation_qa` produces the sharpest Gemini regression (−4.7 pp), consistent with its low Jaccard ($J = 0.333$, Table 7). Last rows: ICREFINE v3 5-task means from Table 3.

	CS-ICL (%)			ProTeGi (%)			Δ (pp)		
Task	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**
CJ	68.6	67.4	67.8	67.4	67.8	65.5	−1.1	+0.4	−2.3
DU	94.3	86.0	90.7	94.7	91.3	88.0	+0.3	**+5.3**	−2.7
DQ	84.7	79.3	81.0	84.3	74.7	79.0	−0.3	−4.7	−2.0
FF	96.7	85.3	83.0	97.7	88.3	83.0	**+1.0**	+3.0	0.0
GS	79.0	68.0	63.0	81.0	68.7	65.7	**+2.0**	+0.7	**+2.7**
SN	93.0	95.8	95.8	92.0	94.4	92.5	−0.9	−1.4	−3.3
Mean (6-task)							+0.2	+0.6	−1.3
ICREFINE v3 (5-task)							−3.4	−1.0	−2.6

GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b. All 3-seed means.
ProTeGi: gpt-4.1-mini, RF scoring, beam size 3, 4 rounds, CS-ICL warm-start; best-by-train selection.

ProTeGi's BBH aggregate (+0.2 pp GPT, +0.6 pp Gemini, −1.3 pp Llama) is less negative than ICREFINE's (−3.4, −1.0, −2.6 pp), likely because gradient-based updates produce smaller, less

Table 17: ProTeGi vs. CS-ICL baseline—3-seed mean RF accuracy on three AGIEval tasks. Δ = ProTeGi $-$ CS-ICL (pp). Best-by-train prompt selection (same as Table 16). GPT-4.1 gains $+3.3$ pp mean, largest on `logiqa_en` ($+8.4$ pp); Gemini is near-zero (-0.5 pp) and Llama slightly positive ($+0.7$ pp).

	CS-ICL (%)			**ProTeGi (%)**			**Δ (pp)**		
Task	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**
lsat_ar	31.3	31.6	29.6	31.0	31.9	31.0	-0.3	$+0.3$	$+1.4$
lsat_lr	88.8	88.6	85.8	90.7	86.8	86.3	$\mathbf{+1.9}$	-1.8	$+0.5$
logiqa_en	45.1	63.5	60.0	53.5	63.6	60.3	$\mathbf{+8.4}$	$+0.1$	$+0.3$
Mean (3-task)							$\mathbf{+3.3}$	-0.5	$+0.7$

GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b. All 3-seed means.
ProTeGi: gpt-4.1-mini, RF scoring, beam size 3, 4 rounds, CS-ICL warm-start; best-by-train selection.

localized artifacts than ICREFINE's explicit case-study generation. The task-structured pattern follows failure geometry: `date_understanding` yields the largest Gemini gain ($+5.3$ pp), consistent with shared arithmetic failure structure; `disambiguation_qa` produces the sharpest Gemini regression (-4.7 pp), consistent with its low source–Gemini Jaccard ($J = 0.333$, Table 7). On AGIEval, GPT-4.1 gains are driven by `logiqa_en` ($+8.4$ pp) and `lsat_lr` ($+1.9$ pp), with `lsat_ar` near-zero (-0.3 pp), giving a $+3.3$ pp mean; Gemini and Llama remain near-zero—the same cross-family asymmetry as ICREFINE on AGIEval. The task-structured alignment between ProTeGi and ICREFINE supports the view that the bottleneck is structural failure-set geometry rather than any feature specific to ICREFINE's distillation operator.

H AGIEval Detailed Results

Table 18 reports full per-task ICREFINE transfer numbers on AGIEval summarised as the "intra-family positive" row in Table 6. Table 19 reports source-model (gpt-4.1-mini) accuracy for ICREFINE and holistic v7 across the AGIEval tasks. Table 20 reports oracle-injected ICREFINE results (Experiment D). Holistic v7 cross-model transfer on AGIEval is reported in Table 15 (Appendix F).

Note on CS-ICL baselines: The GPT-4.1 CS-ICL values in Table 18 and Table 17 (31.3% for `lsat_ar`; 45.1% for `logiqa_en`) reflect the scoring protocol used in those experiments. Subsequent re-measurement with RF scoring on our gpt-4.1-mini-generated cheatsheets yielded substantially higher values (lsat_ar: 70.4%; logiqa_en: 71.5%), consistent with the baselines used in Table 15. The ICREFINE and ProTeGi Δ values in those tables remain internally consistent with their respective baseline protocols.

Table 18: ICREFINE transfer on AGIEval—3-seed mean RF accuracy. CS-ICL: gpt-4.1-mini single-pass warm-start. Δ = ICREFINE $-$ CS-ICL (pp). GPT-4.1 gains strongly ($+3.7$ pp mean), driven by a $+10.6$ pp improvement on `logiqa_en`; Gemini is near-neutral (-0.4 pp) and Llama slightly positive ($+0.7$ pp).

	CS-ICL (%)			**ICREFINE (%)**			**Δ (pp)**		
Task	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**
lsat_ar	31.3	31.6	29.6	31.0	31.0	32.2	-0.3	-0.6	$+2.6$
lsat_lr	88.8	88.6	85.8	89.4	88.1	86.0	$+0.7$	-0.5	$+0.3$
logiqa_en	45.1	63.5	60.0	55.7	63.5	59.2	$\mathbf{+10.6}$	$+0.0$	-0.8
Mean							$\mathbf{+3.7}$	-0.4	$+0.7$

GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b. All 3-seed means.

Table 19: Source-model (gpt-4.1-mini) RF accuracy on AGIEval tasks. ICRefine: 3-seed means (non-RF CS-ICL baseline). Holistic v7: single seed, RF scoring throughout; gpt-4.1-mini test accuracy with best cheatsheet was 64.3% (lsat_ar) and 60.1% (logiqa_en). Direct Δ comparison is not shown as ICRefine and Holistic v7 use different CS-ICL scoring protocols.

	Accuracy (%)		**Δ vs. CS-ICL**
Task	**CS-ICL**	**ICRefine**	**ICRefine**
lsat_ar (floor, ≈28%)	28.1	27.2	-0.9
lsat_lr (ceiling, ≈83%)	83.0	83.7	$+0.7$
logiqa_en (mid, ≈55%)	55.0	54.0	-1.0

Source model: gpt-4.1-mini. ICRefine: cheatsheet_final from 3-seed ICRefine runs.
Holistic v7 source accuracy (RF scoring): lsat_ar 64.3%, logiqa_en 60.1% (single seed).

Table 20: Oracle-injected ICREFINE on AGIEval (Experiment D)—3-seed mean RF accuracy and Δ vs. no-oracle ICREFINE (Table 18). Oracle injection amplifies within-family GPT-4.1 gains by roughly $7\times$ more than cross-family gains; `lsat_lr` (low J) shows no within-family benefit and slight cross-family harm.

	No-oracle (%)			**Oracle (%)**			**Δ (pp)**		
Task	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**	**GPT**	**Gem.**	**Llm.**
lsat_ar	31.0	31.0	32.2	62.9	38.0	38.6	$\mathbf{+31.9}$	$+7.0$	$+6.4$
lsat_lr	89.4	88.1	86.0	90.3	85.6	85.5	$+0.9$	-2.5	-0.5
logiqa_en	55.7	63.5	59.2	72.5	67.0	60.0	$\mathbf{+16.8}$	$+3.5$	$+0.8$
Mean							$\mathbf{+16.5}$	$+2.7$	$+2.2$

GPT = GPT-4.1; Gem. = Gemini-2.0-flash; Llm. = Llama-3.3-70b. All 3-seed means.